

Reads over Atom Spaces: A Mathematical View of Context, Retrieval, and Parameter Routing in Language Models

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Abstract

Attention, external retrieval, and parameter-expert routing all compute query-conditioned weighted reads. We provide a common analytical lens based on valued atom spaces $(\Omega, \mathcal{F}, \nu, \nu)$ and the read operator $R(x; \Omega) = \int_{\Omega} \nu(\omega) d\mu_x(\omega)$, which reduces to a softmax-weighted value combination under a counting measure. First, we derive a Lipschitz-margin sufficient condition for an inserted record to enter top- k retrieval. On 384-dimensional semantic embeddings, the condition is vacuous even when insertion succeeds. Second, we derive exact softmax-tail certificates and an error-budgeted truncation rule. On synthetic corpora with $N = 25,000$, the certificate selects 52–86% of atoms, confirming that it certifies approximation error but is not a sublinear retrieval method. Third, signed reference measures permit exact cancellation of a matched external record by an anti-atom; a 3,000-trial stress test shows that this cancellation is highly sensitive to key perturbations. On CounterFact-500, held-out ROC calibration of a frozen-MiniLM activation threshold ($\theta^* = 0.55$; test AUC 0.829) increases paraphrase activation from 38.2% to 53.2%, while neighborhood false activation rises from 9.6% to 16.1%. Matched-scale GPT-2 experiments place this gated retrieval beside a rank-one parametric reference and GRACE, revealing a generalization–locality trade-off associated with their key spaces. Controlled routing and first-token steering experiments are reported as diagnostics, with their limitations stated explicitly. We do not propose a new editing method or claim superiority; rather, we offer a shared vocabulary and empirical calibrations of what retrieval, metric, and tail arguments can certify.

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